In [119]:

```
# Name: Kachhadiya Janvi
# Enroll No: 202500819010013
# Roll No: IT25A15
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [120]:

```
# Section A Assignment
# Question 1
df = pd.read_csv('studentData.csv')
print(df.head(10))
```

```
   StudentID    Name Gender   Age  Course  Marks  Attendance       Cit
y
0        101    Asha       F    20     BCA     78          85  Ahmedaba
d
1        102    Ravi       M    21     BSc     65          70      Sura
t
2        103    Neha       F    19     BCA     88          90   Vadodar
a
3        104    Aman       M    22   BCom     55          60     Rajko
t
4        105   Pooja       F    20     BSc     92          95  Ahmedaba
d
5        106   Rahul       M    22     BCA     80          85      Delh
i
6        107    Sita       F    21     BSc     72          88    Chenna
i
7        108  Vikram       M    23   BCom     90          91  Bangalor
e
8        109   Priya       F    19     BCA     85          93  Hyderaba
d
9        110   Karan       M    20     BSc     78          80       Pun
e
```

In [121]:

```
# Question 2
print(df.tail(5))
```

```
    StudentID    Name Gender   Age  Course  Marks  Attendance       Ci
ty
10        111   Meera       F    21   BCom     60          70  Ahmedab
ad
11        112    Arun       M    22    BCA     88          92     Mumb
ai
12        113  Anjali       F    20    BSc     77          89    Kolka
ta
13        114   Manoj       M    21   BCom     65          75      Sur
at
14        115   Sneha       F    23    BCA     92          94   Vadoda
ra
```

In [122]:

```
# Question 3
print(df.columns)
```

```
Index(['StudentID', 'Name', 'Gender ', 'Age ', 'Course', 'Marks', 'Att
endance',
       'City'],
      dtype='object')
```

In [123]:

```
# Question 4
df = df.drop(columns=['Attendance'])
print(df)
```

```
    StudentID    Name Gender   Age  Course  Marks       City
0         101    Asha       F    20    BCA     78  Ahmedabad
1         102    Ravi       M    21    BSc     65      Surat
2         103    Neha       F    19    BCA     88   Vadodara
3         104    Aman       M    22   BCom     55     Rajkot
4         105   Pooja       F    20    BSc     92  Ahmedabad
5         106   Rahul       M    22    BCA     80      Delhi
6         107    Sita       F    21    BSc     72    Chennai
7         108  Vikram       M    23   BCom     90  Bangalore
8         109   Priya       F    19    BCA     85  Hyderabad
9         110   Karan       M    20    BSc     78       Pune
10        111   Meera       F    21   BCom     60  Ahmedabad
11        112    Arun       M    22    BCA     88     Mumbai
12        113  Anjali       F    20    BSc     77    Kolkata
13        114   Manoj       M    21   BCom     65      Surat
14        115   Sneha       F    23    BCA     92   Vadodara
```

In [124]:

```
# Question 5
new_df = df[['Name', 'Course', 'Marks']]
print(new_df)
```

```
      Name Course  Marks
0     Asha    BCA     78
1     Ravi    BSc     65
2     Neha    BCA     88
3     Aman   BCom     55
4    Pooja    BSc     92
5    Rahul    BCA     80
6     Sita    BSc     72
7   Vikram   BCom     90
8    Priya    BCA     85
9    Karan    BSc     78
10   Meera   BCom     60
11    Arun    BCA     88
12  Anjali    BSc     77
13   Manoj   BCom     65
14   Sneha    BCA     92
```

In [125]:

```
# Question 6
students_above_80 = new_df[new_df['Marks'] >= 80]
print(students_above_80)
```

```
      Name Course  Marks
2     Neha    BCA     88
4    Pooja    BSc     92
5    Rahul    BCA     80
7   Vikram   BCom     90
8    Priya    BCA     85
11    Arun    BCA     88
14   Sneha    BCA     92
```

In [126]:

```
# Question 7
course_count = df['Course'].value_counts()
print(course_count)
```

```
Course
BCA     6
BSc     5
BCom    4
Name: count, dtype: int64
```

In [127]:

```
# Question 8
missing_values = df.isnull().sum()
print(missing_values)
```

```
StudentID    0
Name         0
Gender       0
Age          0
Course       0
Marks        0
City         0
dtype: int64
```

In [128]:

```
# Question 9
stats_summary = df.describe()
print(stats_summary)
```

```
        StudentID       Age       Marks
count   15.000000  15.000000  15.000000
mean   108.000000  20.933333  77.666667
std      4.472136   1.279881  11.998016
min    101.000000  19.000000  55.000000
25%    104.500000  20.000000  68.500000
50%    108.000000  21.000000  78.000000
75%    111.500000  22.000000  88.000000
max    115.000000  23.000000  92.000000
```

In [129]:

```
# Question 10
sorted_df = df.sort_values(by='Marks', ascending=False)
print(sorted_df)
```

```
    StudentID    Name Gender   Age  Course  Marks       City
4         105   Pooja       F    20     BSc     92  Ahmedabad
14        115   Sneha       F    23     BCA     92   Vadodara
7         108  Vikram       M    23    BCom     90  Bangalore
2         103    Neha       F    19     BCA     88   Vadodara
11        112    Arun       M    22     BCA     88     Mumbai
8         109   Priya       F    19     BCA     85  Hyderabad
5         106   Rahul       M    22     BCA     80      Delhi
0         101    Asha       F    20     BCA     78  Ahmedabad
9         110   Karan       M    20     BSc     78       Pune
12        113  Anjali       F    20     BSc     77    Kolkata
6         107    Sita       F    21     BSc     72    Chennai
1         102    Ravi       M    21     BSc     65      Surat
13        114   Manoj       M    21    BCom     65      Surat
10        111   Meera       F    21    BCom     60  Ahmedabad
3         104    Aman       M    22    BCom     55     Rajkot
```

In [130]:

```
# Section B Assignment
# Question 1
marks_array = np.array([45, 67, 89, 90, 56, 72, 88, 95, 60, 70])
print(marks_array)
```

```
[45 67 89 90 56 72 88 95 60 70]
```

In [131]:

```
# Question 2
print("Maximum Marks:", np.max(marks_array))
print("Minimum Marks:", np.min(marks_array))
print("Average Marks:", np.mean(marks_array))
```

```
Maximum Marks: 95
Minimum Marks: 45
Average Marks: 73.2
```

In [132]:

```
# Question 3
marks_reshaped = marks_array.reshape(2, 5)
print(marks_reshaped)
```

```
[[45 67 89 90 56]
 [72 88 95 60 70]]
```

In [133]:

```
# Question 4
marks_above_70 = marks_array[marks_array > 70]
print(marks_above_70)
```

```
[89 90 72 88 95]
```

In [134]:

```
# Question 5
grace_marks = marks_array + 5
print(grace_marks)
```

```
[ 50  72  94  95  61  77  93 100  65  75]
```

In [135]:

```
# Question 6
sorted_marks = np.sort(marks_array)
print(sorted_marks)
```

```
[45 56 60 67 70 72 88 89 90 95]
```

In [136]:

```
# Question 7
print("Standard Deviation:", np.std(marks_array))
print("Variance:", np.var(marks_array))
```

```
Standard Deviation: 15.942396306703708
Variance: 254.16000000000003
```

In [137]:

```
# Question 8
divisible_by_3 = np.arange(3, 46, 3)
print(divisible_by_3)
```

```
[ 3  6  9 12 15 18 21 24 27 30 33 36 39 42 45]
```

In [138]:

```
# Question 9
random_numbers = np.random.randint(10, 101, 10)
print(random_numbers)
```

```
[ 93  95  42  83  56  95  87 100  47  57]
```

In [139]:

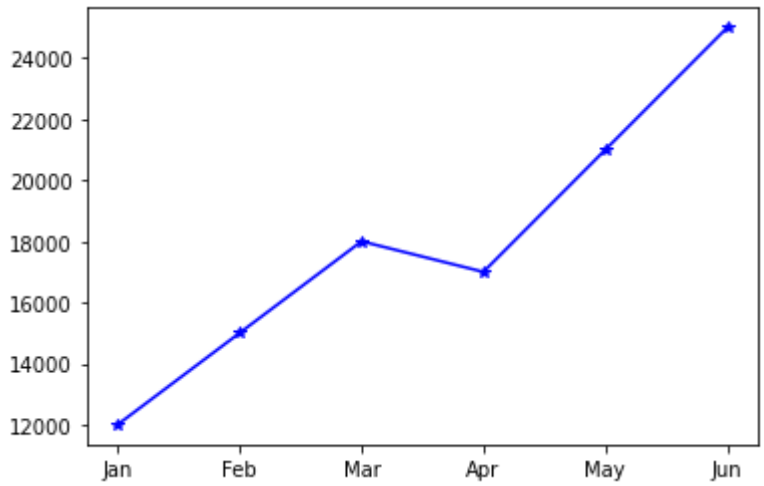
```
# Question 10
marks_replaced = np.where(marks_array < 60, 60, marks_array)
print(marks_replaced)
```

```
[60 67 89 90 60 72 88 95 60 70]
```

In [140]:

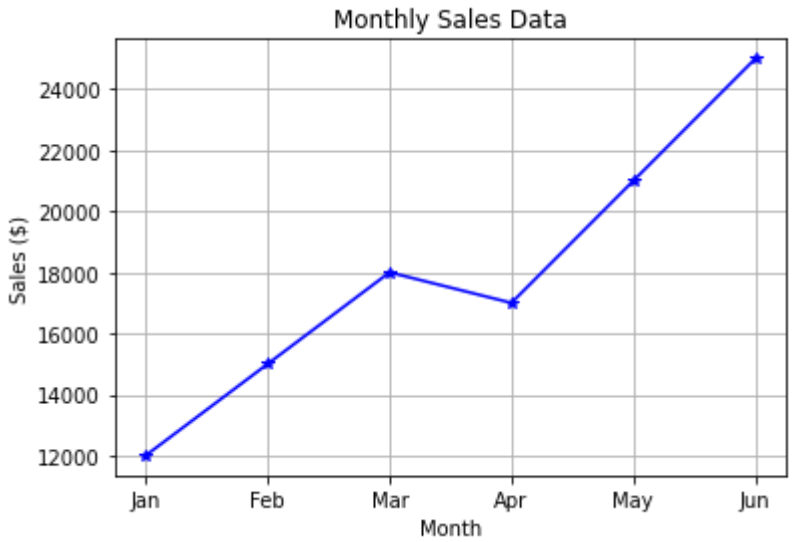
```
# Section C Assignment
# Question 1
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [12000, 15000, 18000, 17000, 21000, 25000]

plt.plot(months, sales, marker='*', linestyle='-', color='b')
plt.show()
```



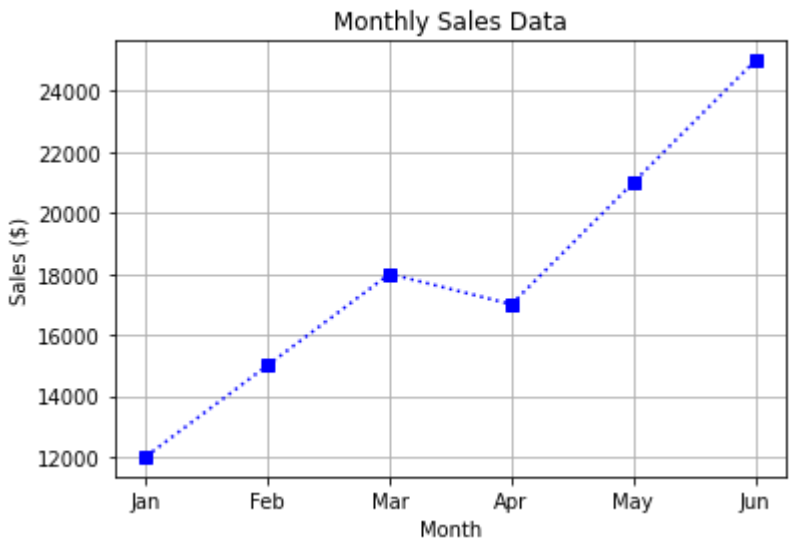
In [141]:

```
# Question 2
plt.plot(months, sales, marker='*', linestyle='-', color='b')
plt.title('Monthly Sales Data')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid()
```



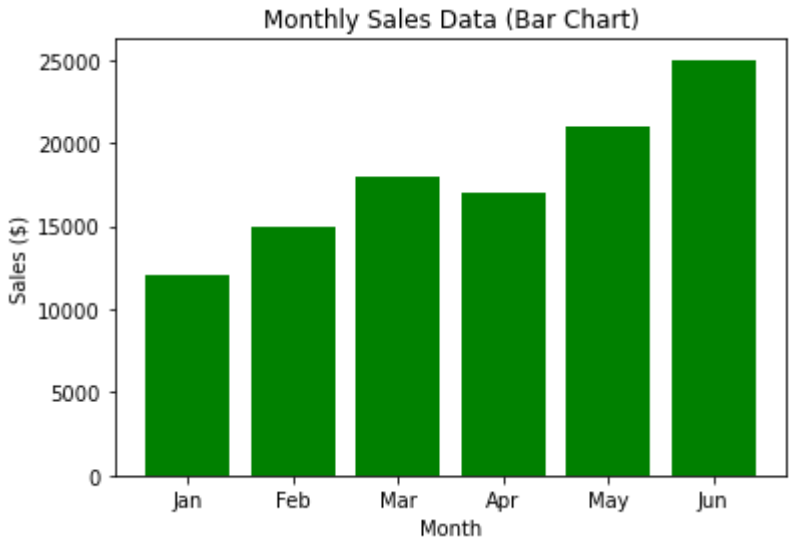
In [142]:

```
# Question 3
plt.plot(months, sales, marker='s', linestyle=':', color='b')
plt.title('Monthly Sales Data')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid()
```



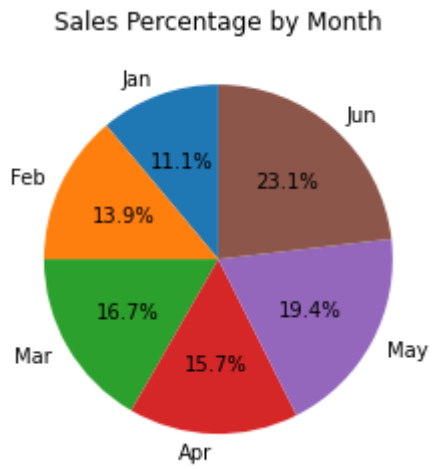
In [143]:

```
# Question 4
plt.bar(months, sales, color='green')
plt.title('Monthly Sales Data (Bar Chart)')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.show()
```



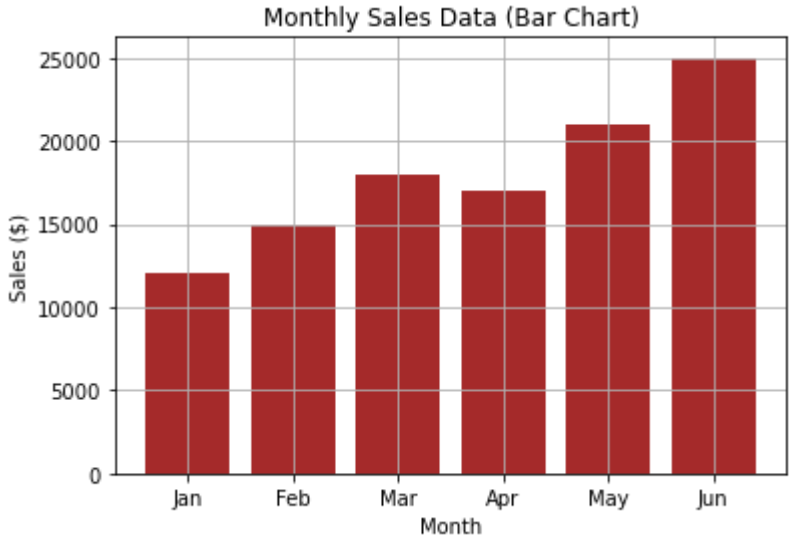
In [144]:

```
# Question 5
plt.pie(sales, labels=months, autopct='%1.1f%%', startangle=90)
plt.title('Sales Percentage by Month')
plt.show()
```



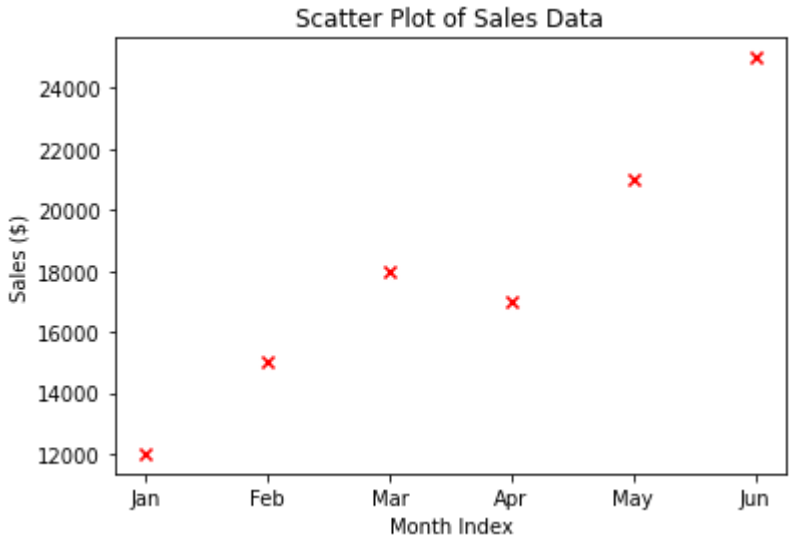
In [145]:

```
# Question 6
plt.bar(months, sales, color='brown')
plt.title('Monthly Sales Data (Bar Chart)')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid()
```



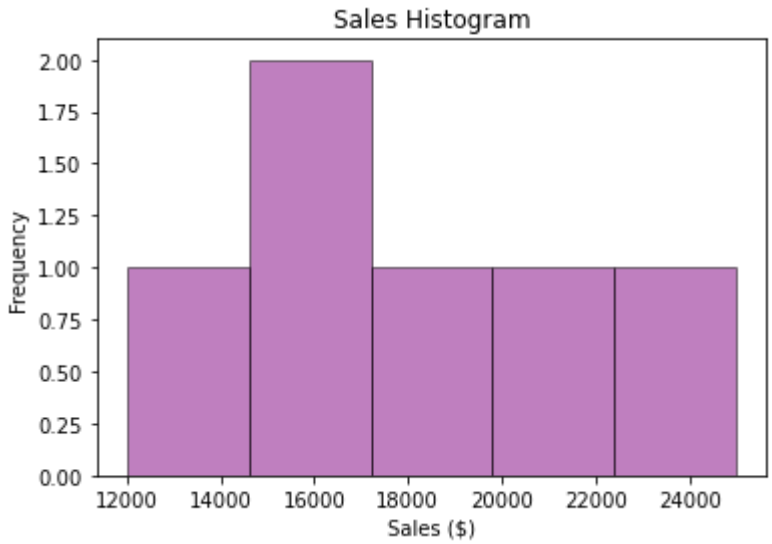
In [148]:

```
# Question 7
plt.scatter(months, sales, color='r', marker='x')
plt.title('Scatter Plot of Sales Data')
plt.xlabel('Month Index')
plt.ylabel('Sales ($)')
plt.show()
```



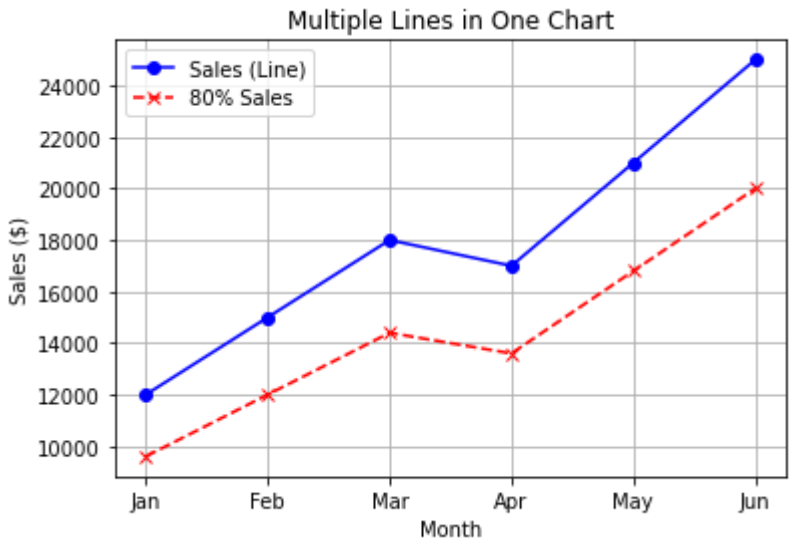
In [147]:

```
# Question 8
plt.hist(sales, bins=5, color='purple', edgecolor='black', alpha=0.5)
plt.title('Sales Histogram')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.show()
```



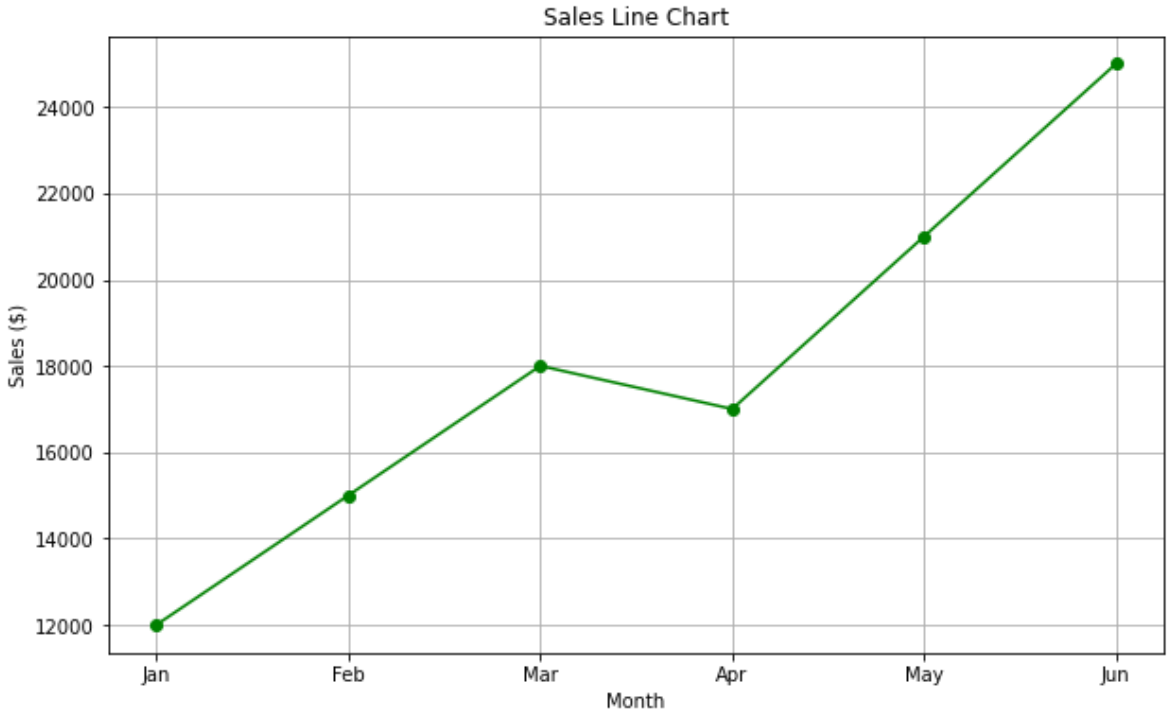
In [149]:

```
# Question 9
plt.plot(months, sales, marker='o', label='Sales (Line)', color='b')
plt.plot(months, [x * 0.8 for x in sales], marker='x', label='80% Sales', color='r'
plt.title('Multiple Lines in One Chart')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.legend()
plt.grid()
```



In [79]:

```
# Question 10
plt.figure(figsize=(10, 6))
plt.plot(months, sales, marker='o', color='g', label='Sales')
plt.title('Sales Line Chart')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid()
plt.savefig('sales_chart.jpg')
```



In [150]:

```
# Section D Assignment
# Question 1
df = pd.read_csv('categoryData.csv')
print(df)
```

```
   Month     Category  Sales
0    Jan  Electronics  15000
1    Feb     Clothing  10000
2    Mar  Electronics  17000
3    Apr    Furniture  13000
4    May     Clothing  12000
5    Jun  Electronics  20000
6    Jul    Furniture  14000
7    Aug  Electronics  22000
8    Sep    Furniture  15000
9    Oct     Clothing  11000
10   Nov  Electronics  18000
11   Dec    Furniture  16000
```

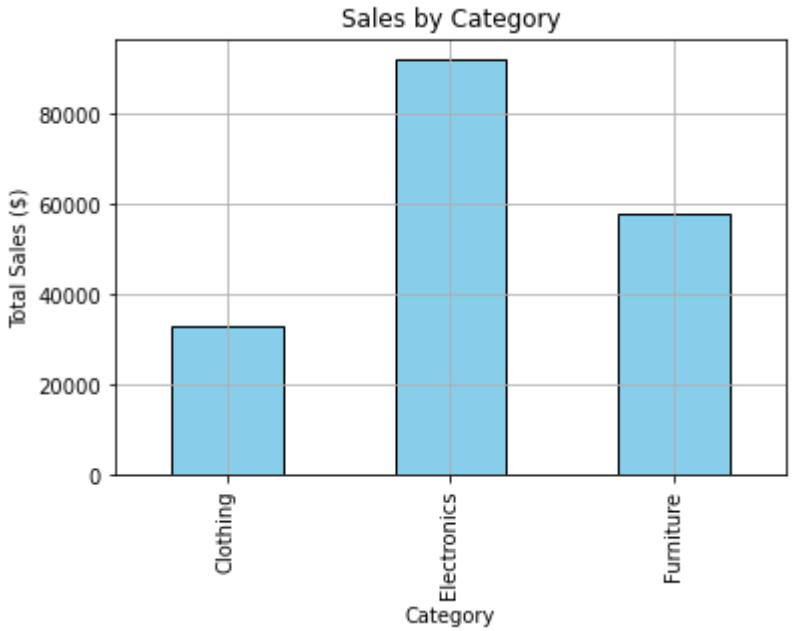
In [151]:

```
# Question 2
grouped_data = df.groupby('Category')['Sales'].sum()
print(grouped_data)
```

```
Category
Clothing       33000
Electronics    92000
Furniture      58000
Name: Sales, dtype: int64
```

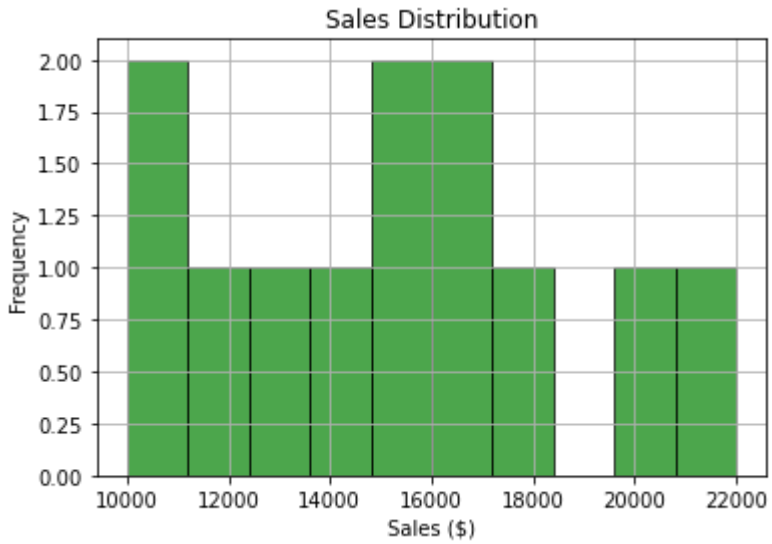
In [152]:

```
# Question 3
grouped_data.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales ($)')
plt.grid()
```



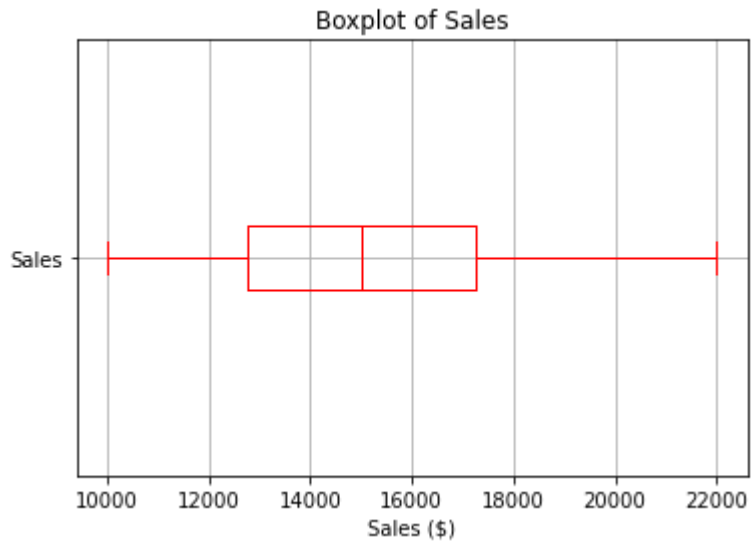
In [153]:

```
# Question 4
df['Sales'].plot(kind='hist', bins=10, color='green', alpha=0.7, edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.grid()
```



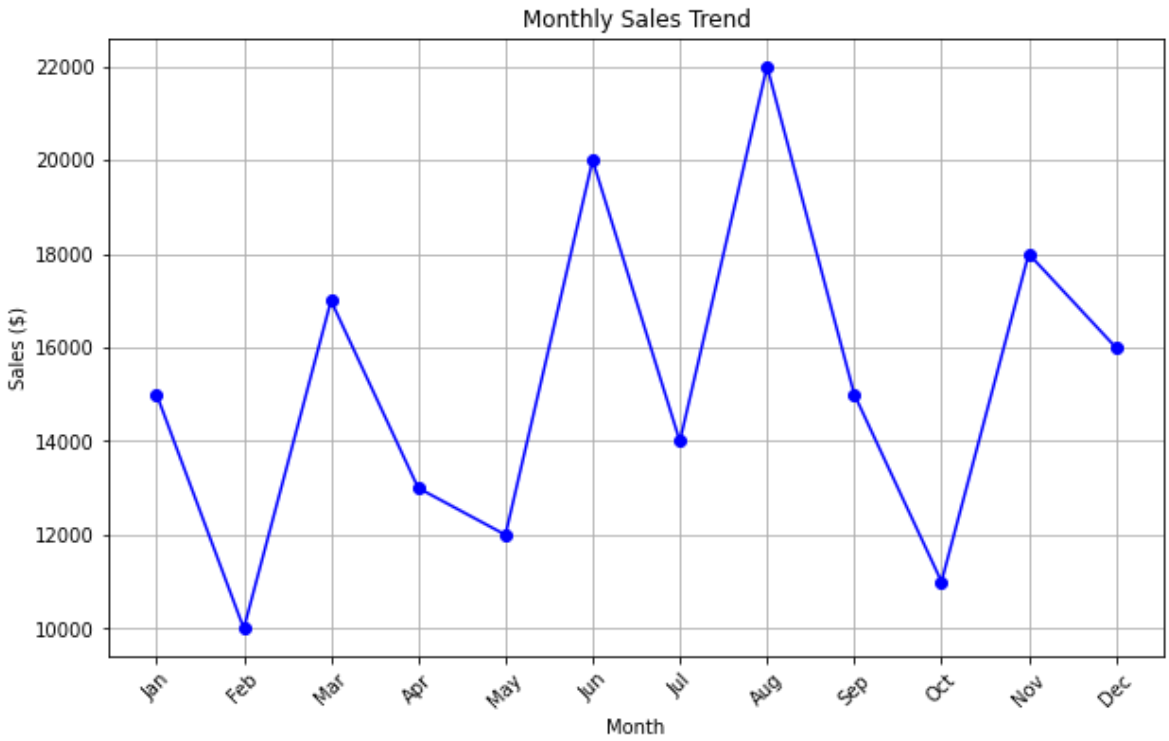
In [154]:

```
# Question 5
df['Sales'].plot(kind='box', color='red', vert=False)
plt.title('Boxplot of Sales')
plt.xlabel('Sales ($)')
plt.grid()
```



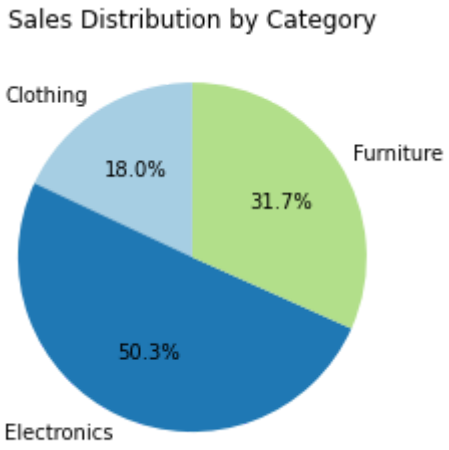
In [155]:

```
# Question 6
month_array = df['Month'].values
sales_array = df['Sales'].values
plt.figure(figsize=(10, 6))
plt.plot(month_array, sales_array, marker='o', color='blue')
plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.xticks(rotation=45)
plt.grid()
```



In [156]:

```
# Question 7
grouped_data.plot(kind='pie', autopct='%1.1f%%', startangle=90, colors=plt.cm.Paire
plt.title('Sales Distribution by Category')
plt.ylabel('')
plt.show()
```



In [ ]:

```

```